Week 4 Lecture Activity

Write an Abstract Machine called Favourites with two sets, **PEOPLE** and **COLOR**. Create a relation named **favColor** between PEOPLE and COLOR and initialize it with some values.

Implement the following operations:

1. addFavourite(person, color) – Adds a person's favourite color to the relation.
2. removeFavourites(person, color) – Removes a specific person–color pair from the relation.
3. removeAllFavourites(person) – Removes all favourite colors associated with the specified person.
4. getFavouriteColor(person) – Returns the favourite color or colors associated with the specified person.
5. getPeople(color) – Returns the people associated with the specified color.
6. updateFavourites(person, color) – Updates the specified person's favourite color to the given color, replacing their existing color association if necessary.
7. getPeopleWithColors(colors)- Returns all the maplets (person and color) associated with specific color